



NYU

TANDON SCHOOL
OF ENGINEERING

NYU TANDON CAREER HUB · FALL 2026

Data Science Bootcamp

Week 2 · Numpy & Pandas

Instructor: **Angelin Ann Jacob**

Section 1 · Tue Oct 6

Section 2 · Thu Oct 8

Useful Links

Zoom

All sessions are virtual. Use **ONLY your NYU SSO Login to be considered for attendance**

Brightspace

Assignments, submissions and the discussion board, monitored Monday to Thursday.

GitHub

Slides and Colab notebooks for every session.

Email

datasciencebootcamp@nyu.edu for questions, extensions and attendance approvals.

1:1 coaching

Optional 30-minute technical interview prep. [Book it through Handshake.](#)



NYU

TANDON SCHOOL
OF ENGINEERING

Numpy & Pandas

Get to the libraries used in your day to day tasks as a Data Specialist!

**NYU**TANDON SCHOOL
OF ENGINEERING

NYC Open Data

<https://www.nyc.gov/opendata>

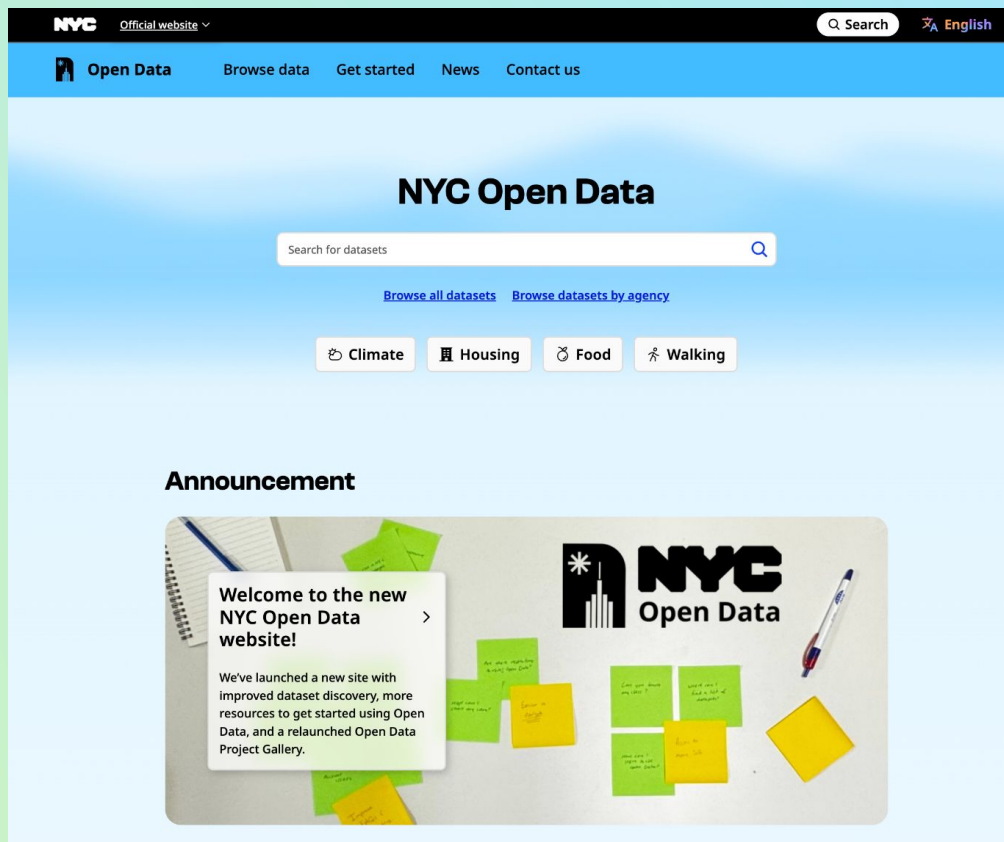
Dataset name:

Brooklyn Bridge Automated Pedestrian Counts
Demonstration Project (Historic)

Data Provided By:

Department of Transportation (DOT)

Visit: [NYC Open Data - City of New York - Brooklyn Bridge Automation](#)





HOW TO READ TODAY'S SESSION

Two parts for every code block

THE STORY

The question about the bridge

What we want to know, in plain English.

When do people cross? Which direction dominates? Does rain matter?



THE TOOL

The library and function

How we get the answer, in code.

```
np.reshape()    pd.read_csv()  
df.groupby()    df.sort_values()
```

EXAMPLE

Story: Which hours of the day are busiest?



```
bridge.groupby("hour")  
    .pedestrians.mean()
```



PART 1 · NUMPY · INDEXING & SLICING

Pick out exactly what you need

0	1	2	3	4
44	30	25	20	18
-5	-4	-3	-2	-1

`pedestrians[:5]`

Five real counts, two ways to number them: from the front, or from the back.

Spot-check random hours with `np.random.randint`.

THE STORY

Which hour was recorded first?
Which last?

THE TOOL

```
pedestrians[0]  
pedestrians[-1]  
direction[0, 0]
```

THE STORY

Pull out one full day, or the same
hour every day.

THE TOOL

```
a[:24]      # one day  
a[:, :24]   # every 24th  
a[:, :-1]   # newest first
```

THE STORY

Edit a slice. Does the original
change?

THE TOOL

```
view = a[:5]  
view[0] = -1 # a too?!  
a[:5].copy()
```



PART 1 · NUMPY · RESHAPING & AXES

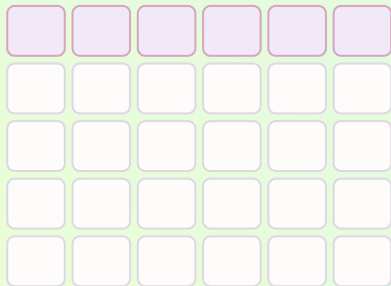
Turn hours into days



7,296 hourly counts in one row

↓ `reshape((304, 24))`

24 hours →



304 days ↓

axis=0

collapse the days: one
typical day

axis=1

collapse the hours: one
total per day

THE STORY

Which hours are busiest on a typical day?

One row per day, one column per hour.

Encoding columns and rows with positional info!

THE TOOL

```
grid = ped.reshape((304, 24))
grid.mean(axis=0)
grid.sum(axis=1)
typical_day[np.newaxis, :]
np.where(cond, "busy", "quieter")
```



PART 2 · PANDAS · THE TABLE

Meet the table

THE STORY

What did we just load? How big is it?

Typical values. Is the mean far from the median?

How many people in total? How common is each weather?

Busiest hours? Rainy versus dry? The single busiest hour?

THE TOOL

```
bridge.head()  bridge.tail()  
bridge.shape  bridge.info()
```

```
bridge.describe()  
bridge.dtypes
```

```
bridge["Pedestrians"].sum()  
bridge["weather_summary"].value_counts()
```

```
bridge.sort_values(ascending=False)  
bridge[bridge["precipitation"] > 0]  
bridge.loc[bridge["Pedestrians"].idxmax()]
```



PART 2 · PANDAS · GROUPING

Group, compare, summarize

Every hourly row

one row = one hour



```
analysis.groupby("hour")
```

split into 24 piles, one per hour



```
.pedestrians.mean()
```

one number per pile: a Series

THE STORY

Tidy names, then pull out the hour.

THE TOOL

```
df.rename(columns={...})  
df.pedestrians  
df["time"].dt.hour
```

THE STORY

Mean, min and max for every weather label.

THE TOOL

```
.groupby("weather_summary")  
.agg([...])  
.describe()
```

THE STORY

Readable labels, and tables we build ourselves.

THE TOOL

```
.apply(lambda q: ...)  
pd.Series(...)  
pd.DataFrame({...})
```



NYU

TANDON SCHOOL
OF ENGINEERING

Before next week

Take-home

The exercises at the end of today's notebook. Submit on Brightspace within one week, by 11:59 PM.

Feedback form

Fill it out after today's session. It counts toward your credential.

Week 3: SQL Fundamentals

Work with Databases and get familiar with SQL Basics!

See you at the social: **Mon, Oct 5, 3–5 PM** · 2 MetroTech Center, Room 1054